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Modeling the functioning of rubber trees (*Hevea brasiliensis*) under changing climate and management-driven disturbances

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Rubber Research Station in Chachoengsao, Thailand.

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Hevea brasiliensis, a strategic crop under threat?

🔍 Two major threats to rubber cultivation in Southeast Asia

1.Climate change:

- Rising average air temperatures, the most critical threat to rubber plantations ?
- Water stress limits growth, prolonging the immature phase and reducing potential yield *(Thaler, 2021)*

2.Declining agricultural labor availability

FORSEA « Forecasting impacts of climate change and workforce availability on natural rubber commodity chain in South-East Asia »

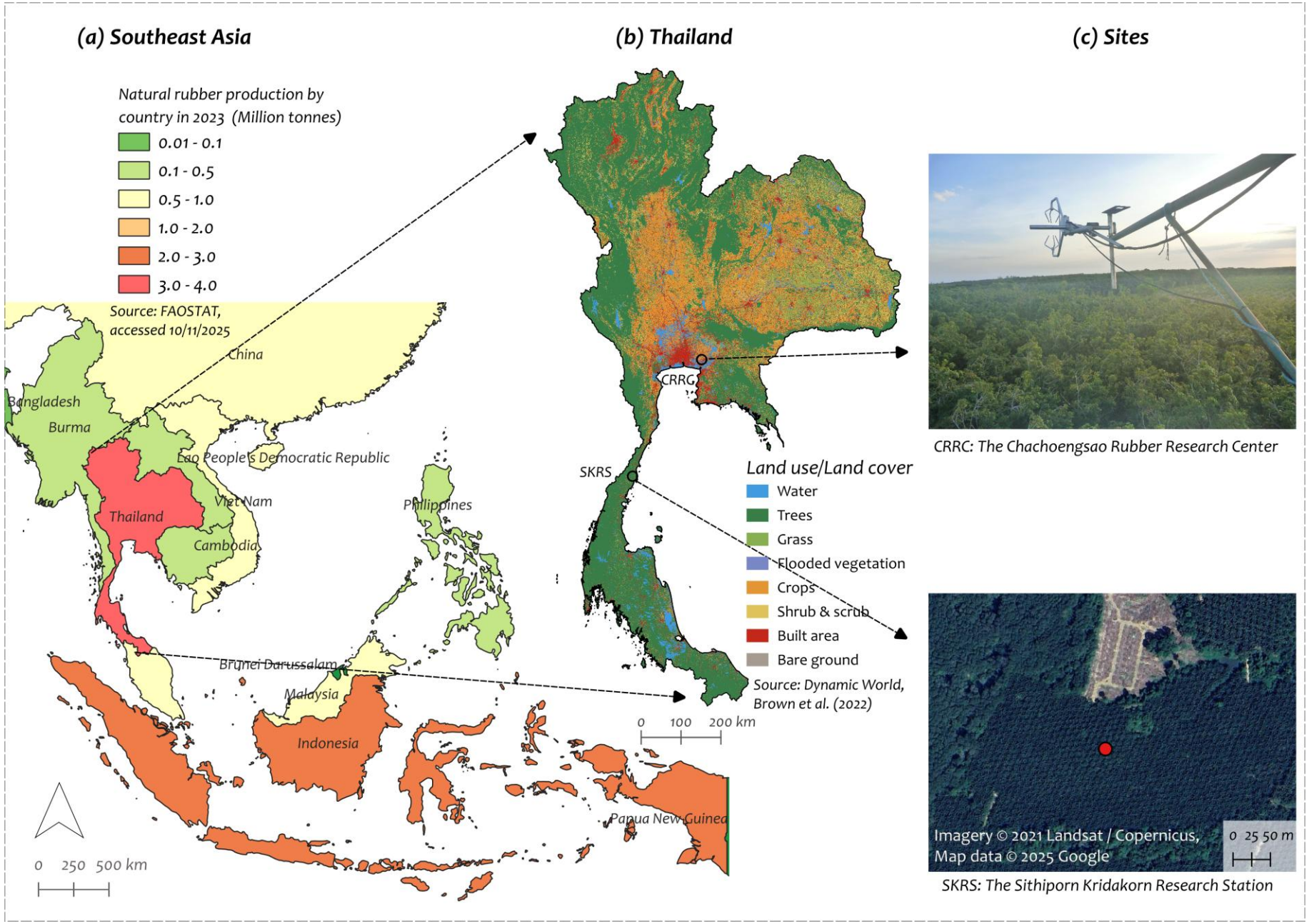
Research objectives

- **Adapt CASTANEA-MAESPA, a model designed to simulate tree plantations, based on ecophysiological functions. We started from a version adapted to eucalyptus plantation.**

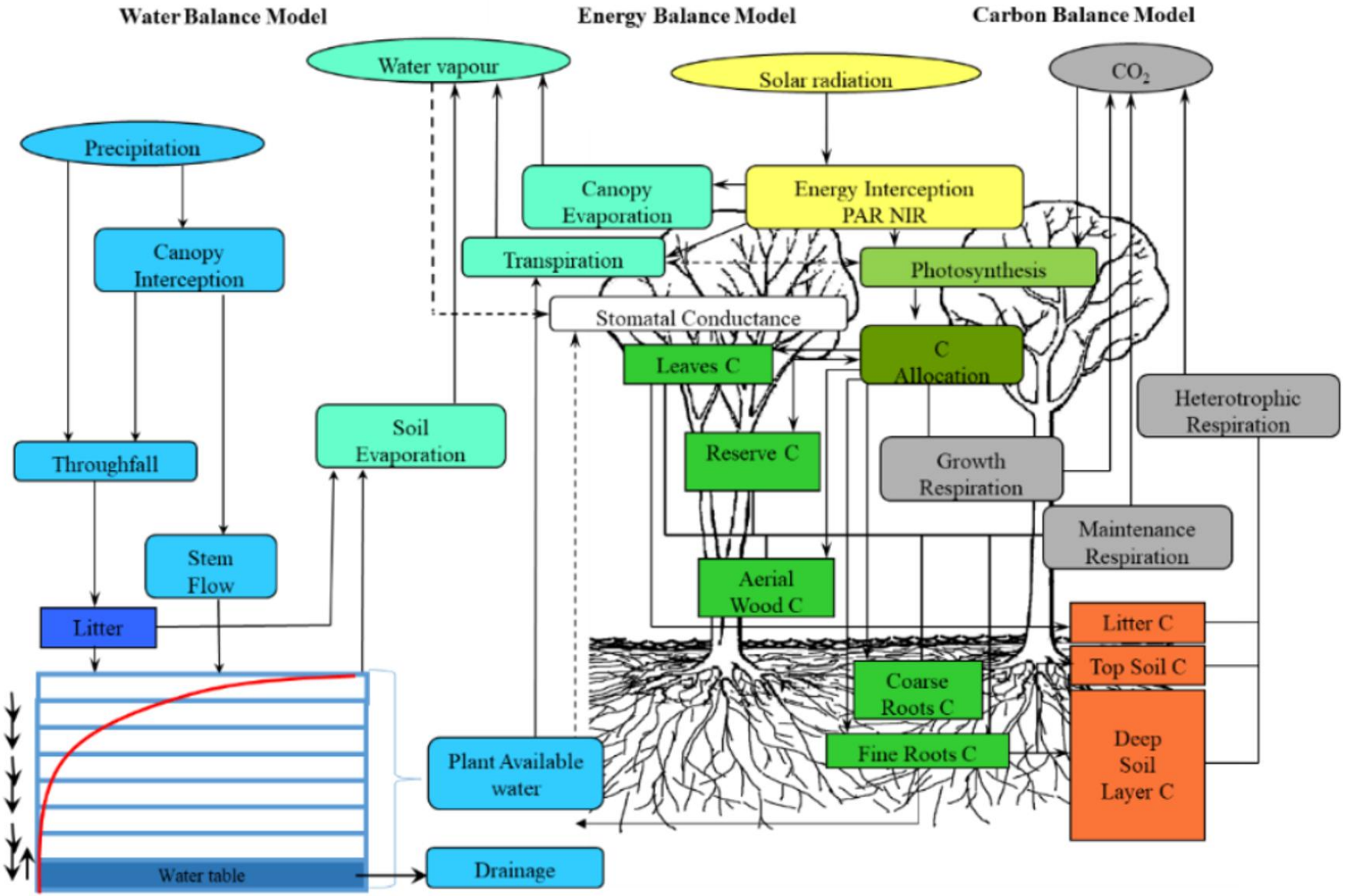
- **Develop a specific module simulating the latex production processes and integrate it into CASTANEA-MAESPA.**

- **Test the impact of different climate change scenarios on biomass and latex productivity in rubber tree plantations.**

Study area


(a) Southeast Asia
(b) Thailand
(c) Sites
Natural rubber production by country in 2023 (Million tonnes)
0.01 - 0.1
0.1 - 0.5
0.5 - 1.0
1.0 - 2.0
2.0 - 3.0
3.0 - 4.0
Source: FAOSTAT, accessed 10/11/2025
China
Bangladesh
Burma
Lao People's Democratic Republic
Thailand
Viet Nam
Cambodia
Philippines
Brunei Darussalam
Malaysia
Indonesia
Papua New Guinea
0 250 500 km
CRRG
SKRS
Land use/Land cover
Water
Trees
Grass
Flooded vegetation
Crops
Shrub & scrub
Built area
Bare ground
Source: Dynamic World, Brown et al. (2022)
0 100 200 km
CRRC: The Chachoengsao Rubber Research Center
Imagery © 2021 Landsat / Copernicus,
Map data © 2025 Google
0 25 50 m
SKRS: The Sithiporn Kridakorn Research Station

Methods



The CASTANEA-MAESPA model Source: Le Maire et al. (2023)

Climate data

Sources of input weather → POWER Nasa | EC tower CRRC | CMIP6 Projections (SSPs)

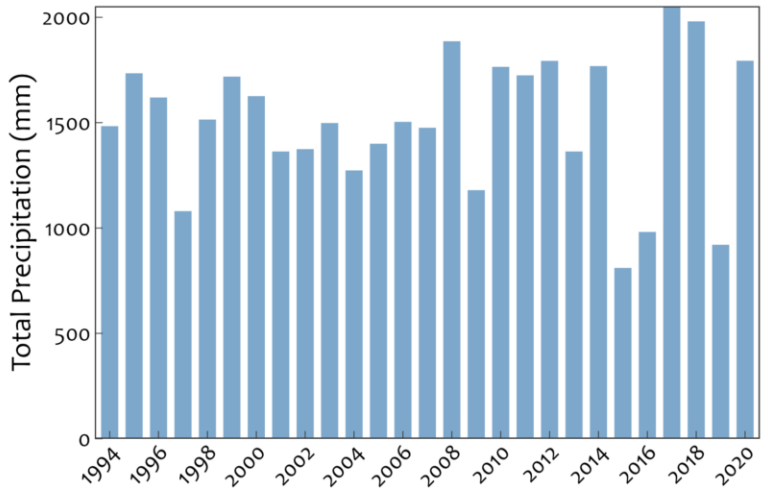
Simulation plan SKRS → year: 2018 … 2021

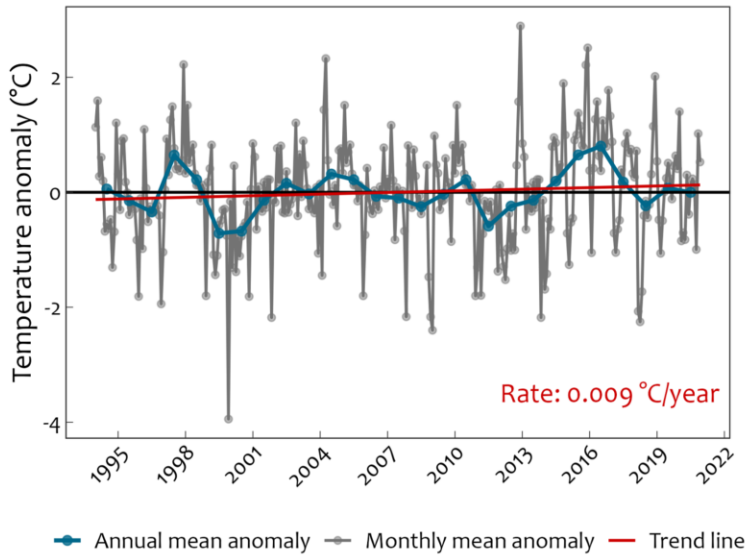
Simulation plan CRRC → year: 1994 … 2015 … 2018 2019 2020 … 2045 2046 … 2071

model calibration period

26 years rotations: 1st 2nd 3rd

CRRC





Input MIROC6 | Input MPI-ESM1-2-LR | Input AWI-CM-1-1-MR

→ CASTANEA

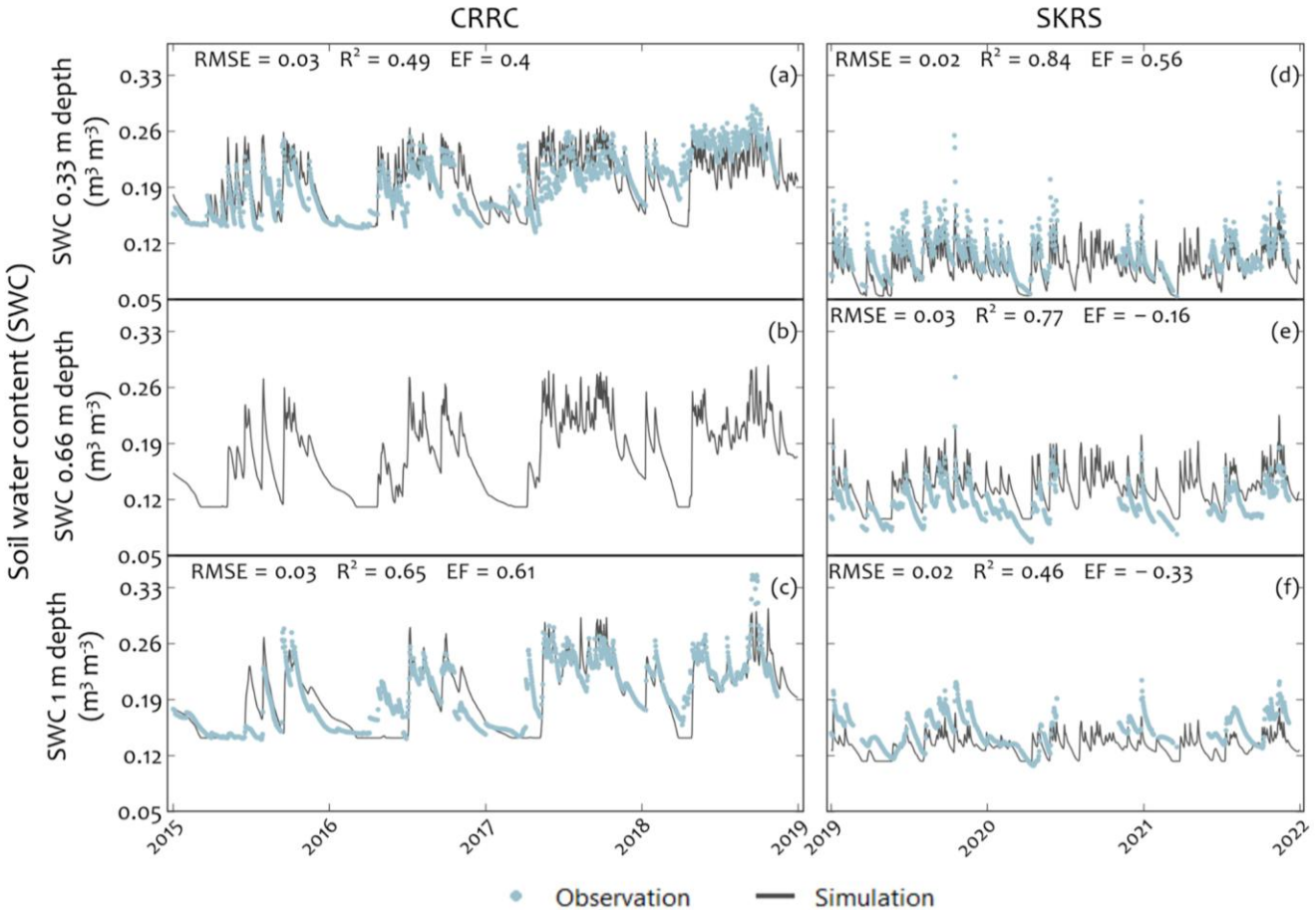
→ Output MIROC6 | Output MPI-ESM1-2-LR | Output AWI-CM-1-1-MR

→ Average CASTANEA outputs

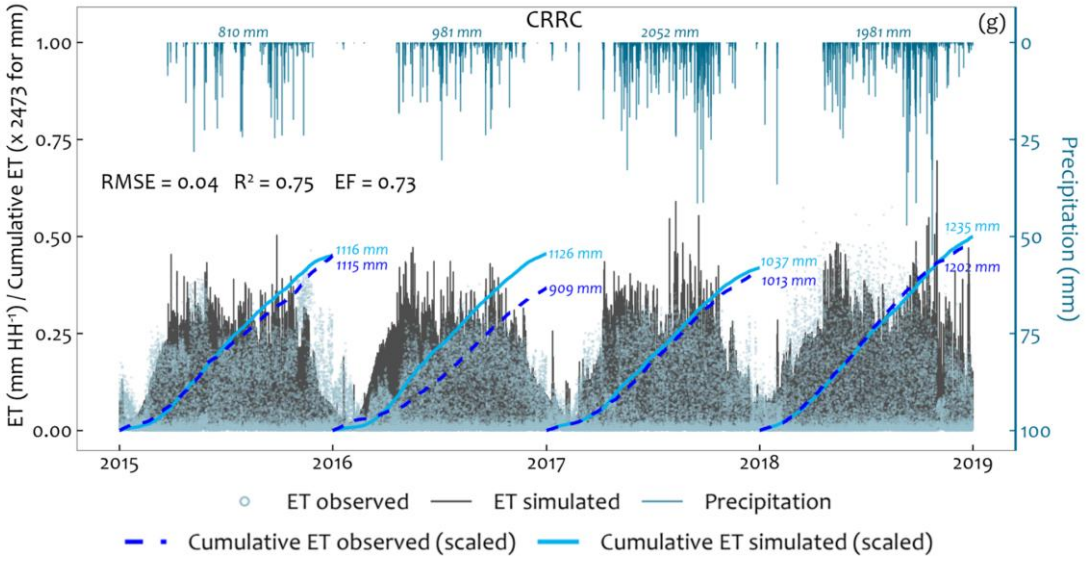
From 1994 to 2020 at CRRC:

- *Variability in annual precipitation*
- *Slight increase in monthly mean temperature anomalies*

Water balance of the rubber plantation



SWC simulations agreed well with observations at CRRC, particularly in layer 3, while at SKRS, performance was best in layer 1 and decreased with depth.



Evapotranspiration (ET) simulations at CRRC showed strong agreement at the half-hourly scale and reproduced annual cumulative ET well in most years, although performance was overestimated in 2016.

Rubber tree, a deciduous species with rhythmic growth

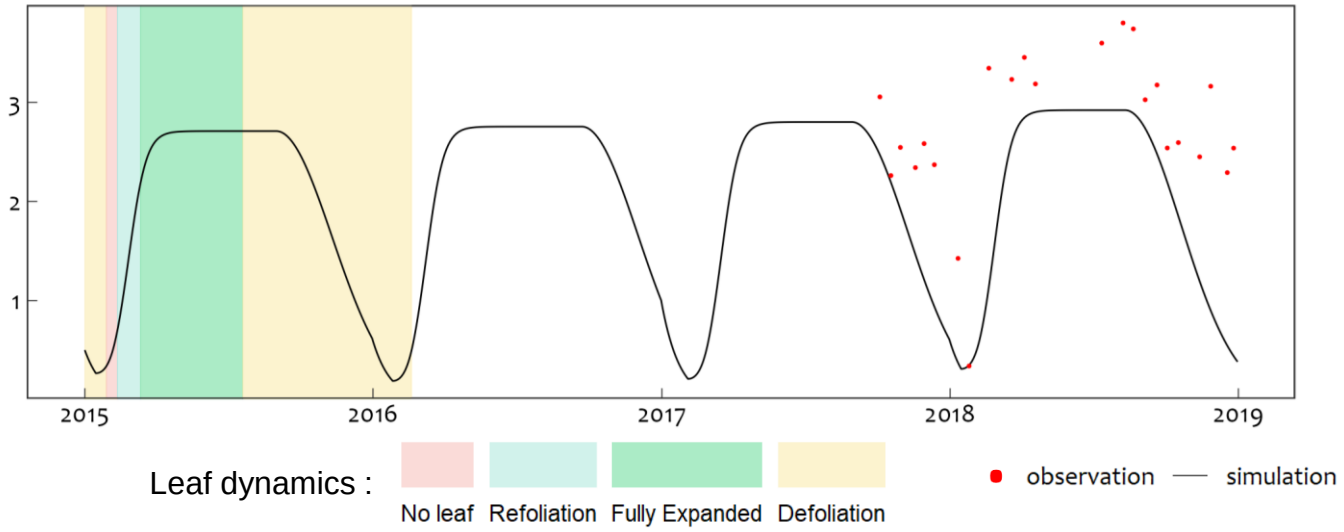
Philippe Thaler ©



No leaf (1)	Refoliation (2-4)	Fully expanded (5-8)	Defoliation (9-12)

Leaf dynamics (months)*

Leaf area index ($m^2/m^2/d$)



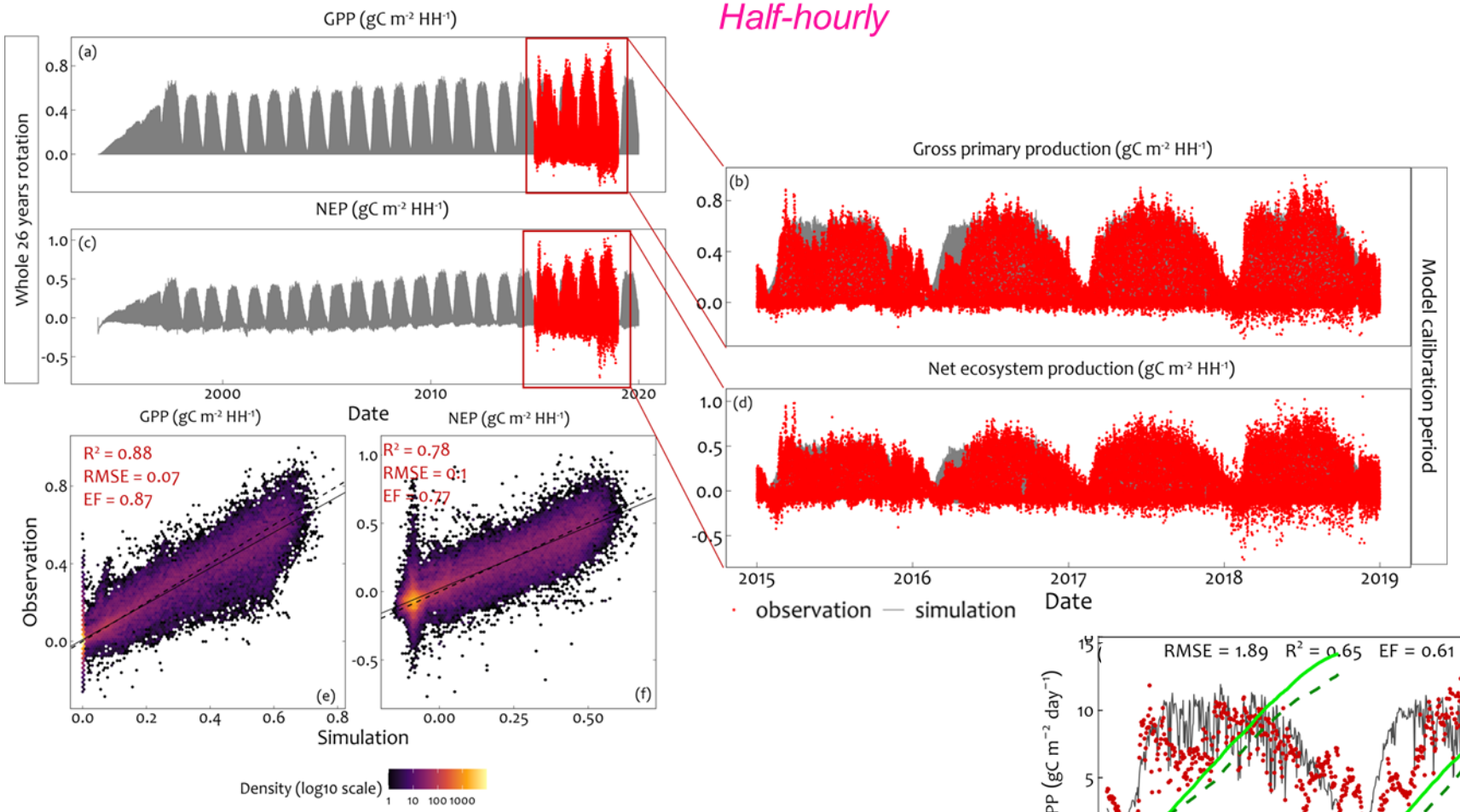
Model Assumptions:

- Refoliation: Leaf production is triggered by a cumulative low-temperature index and also depend on height.
- Leaf expansion: Leaf area and mass expand following sigmoid functions.
- Leaf defoliation: Senescence is triggered by a cumulative low-temperature index, with leaf loss following exponential decay until complete defoliation. Seasonal dynamics are activated from the third year of the rotation onward

The model simulates a good LAI dynamic, however the observations and simulations are not a perfect match. More data is needed to improve the simulation of leaf dynamics

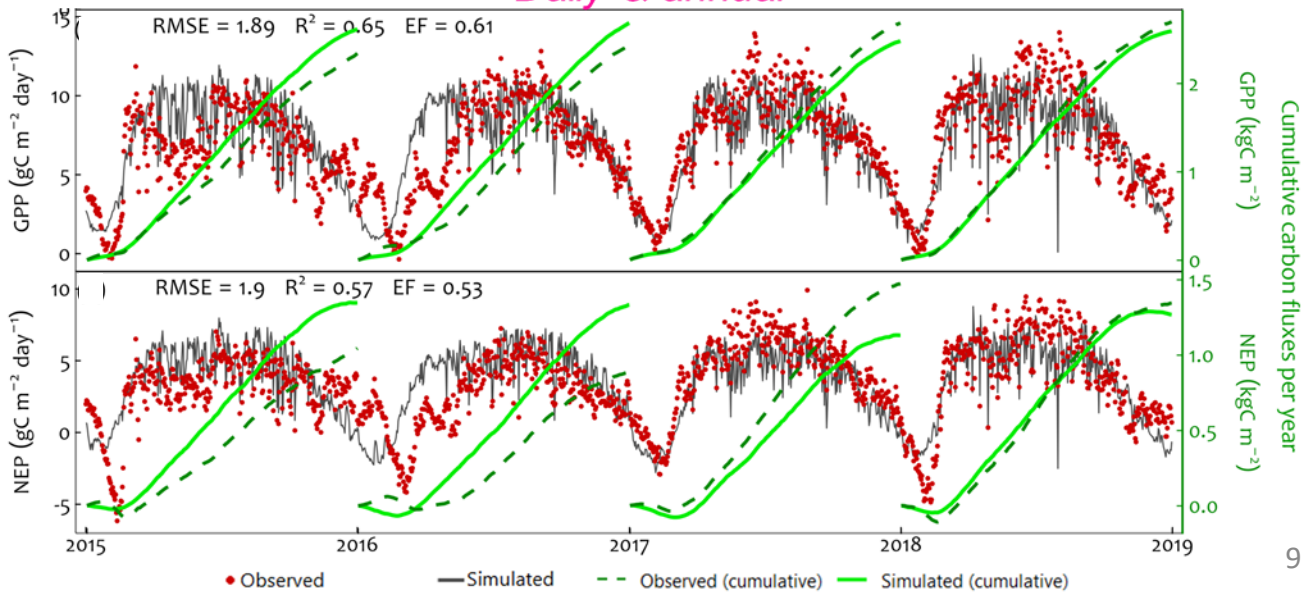
Carbon fluxes (GPP & NEP) of the rubber plantation

Half-hourly



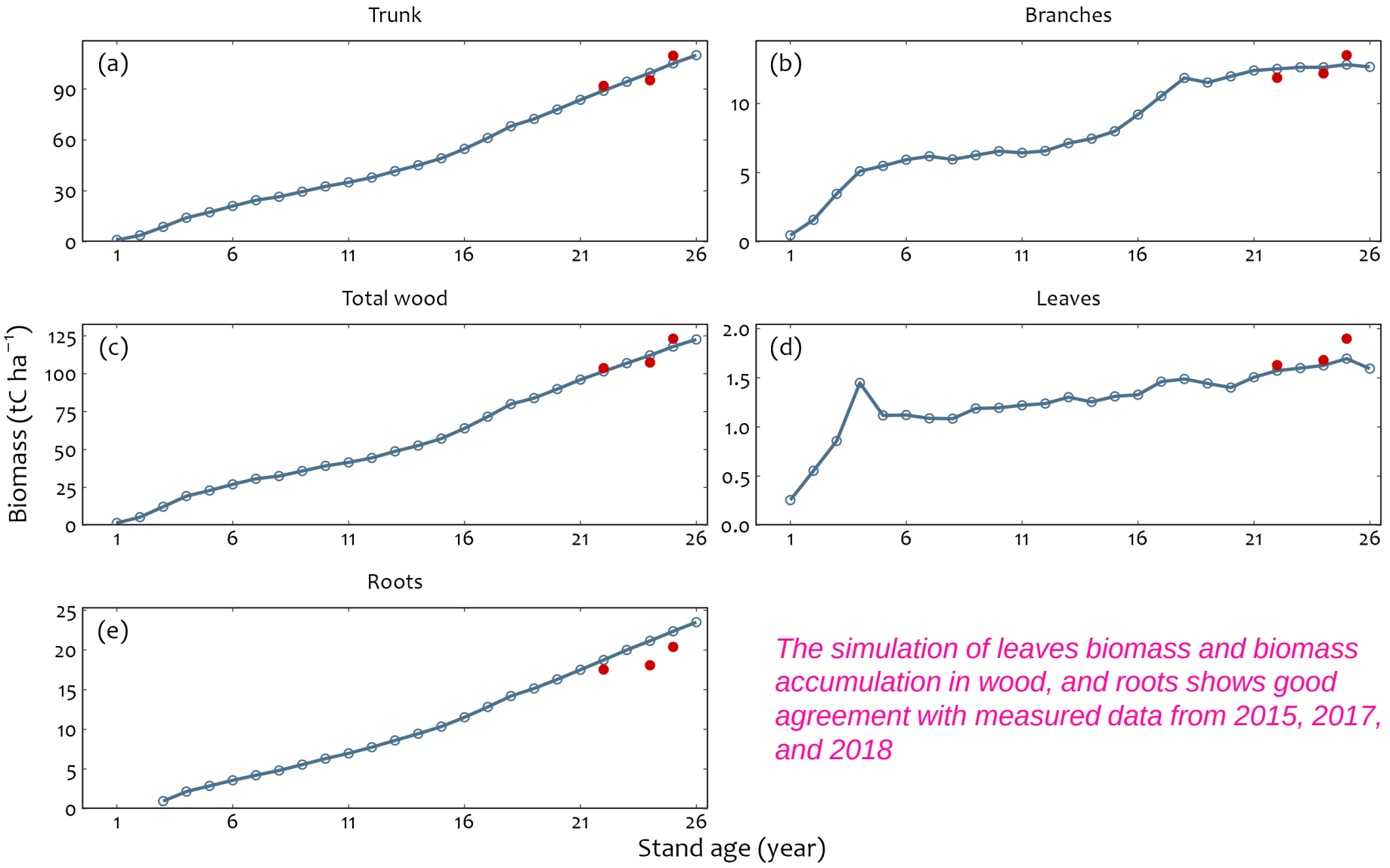
Daily and annual GPP and NEP: *The model generally reproduced daily and annual carbon fluxes, with moderate daily performance for GPP (EF=0.61) and NEP (EF=0.53), despite some seasonal overestimations/underestimates and annual discrepancies.*

Daily & annual



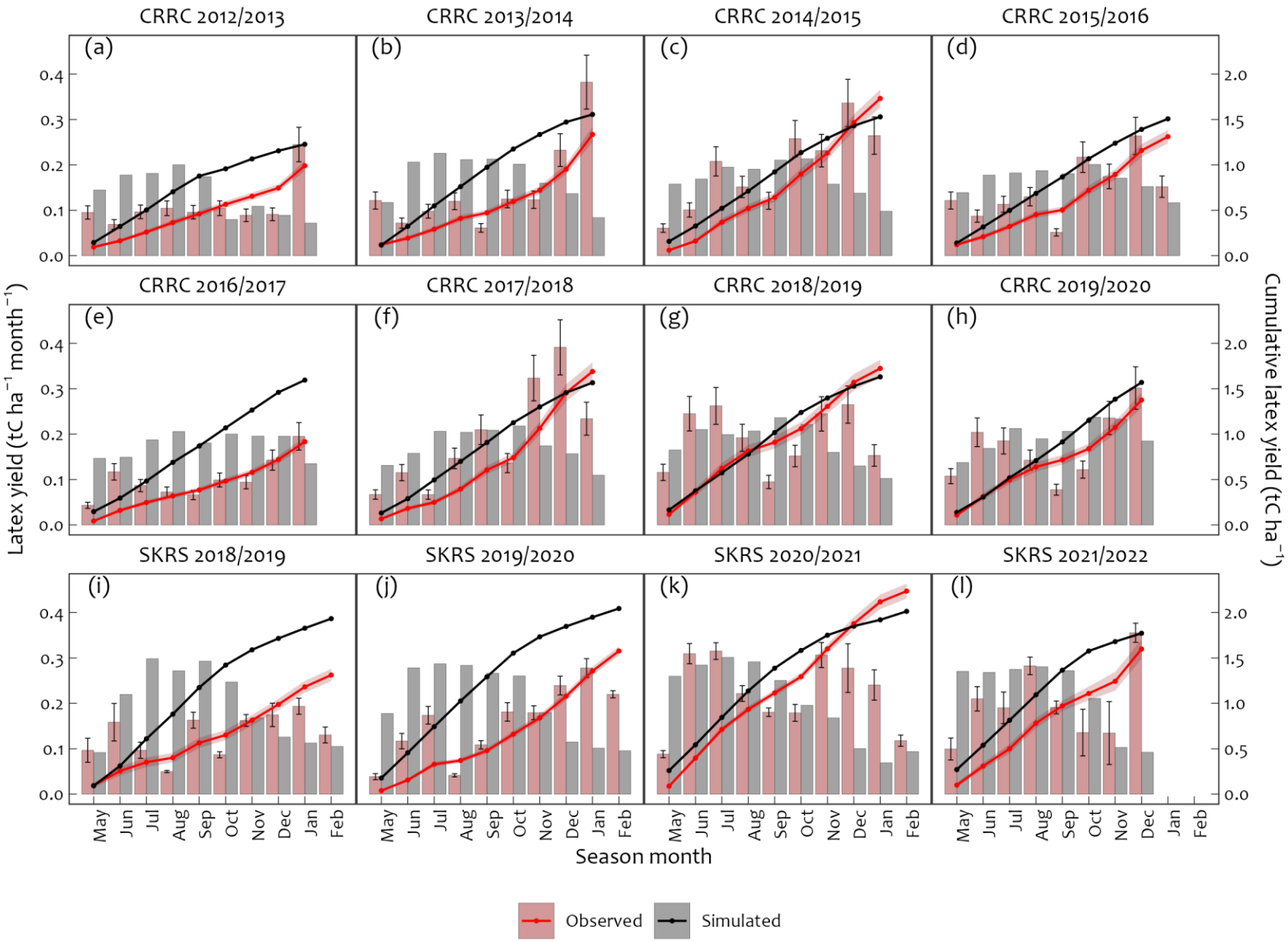
Half-hourly GPP and NEP: *The model captured well short-term fluctuations, with strong performance for GPP (EF=0.87) and NEP (EF=0.77).*

Biomass accumulation in rubber tree



The simulation of leaves biomass and biomass accumulation in wood, and roots shows good agreement with measured data from 2015, 2017, and 2018

Latex yield simulation



Note: The model simulates potential production, without accounting for losses due to operational constraints, management issues, tree damage, or extreme events.

CRRC (calibration site): *The model reasonably reproduced annual latex yields (average: 1.52 vs. 1.38 t C ha^{-1} yr^{-1} measured) but captured monthly variability moderately (R^2 = 0.35), with seasonal over- and underestimations.*

SKRS (validation site): *The model generally overestimated annual yields (average: 1.94 vs. 1.68 t C ha^{-1} yr^{-1} measured) and showed limited ability to reproduce monthly variability (R^2 = 0.13)*

Full carbon balance of the rubber plantation (e.g. 2018)

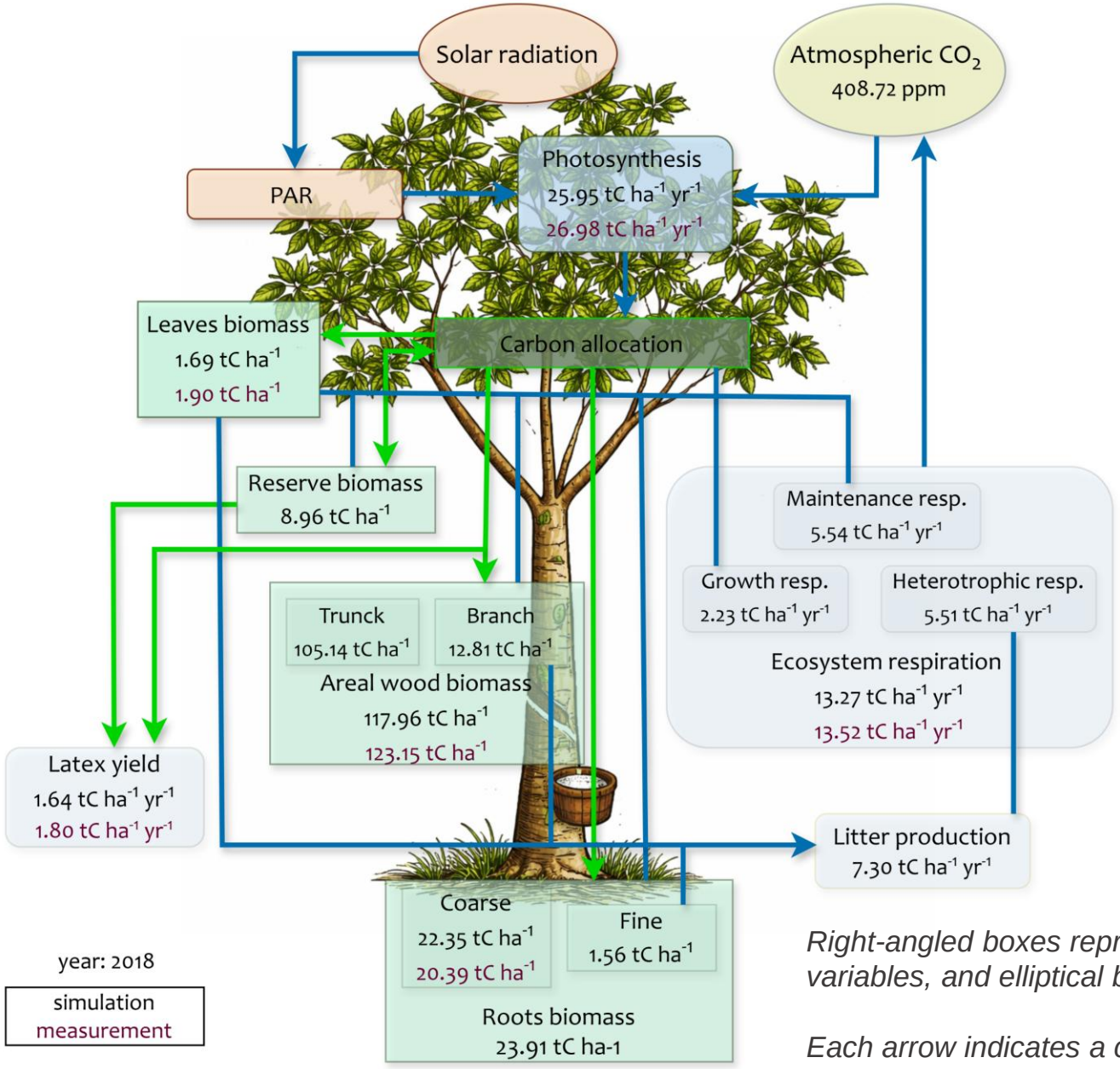
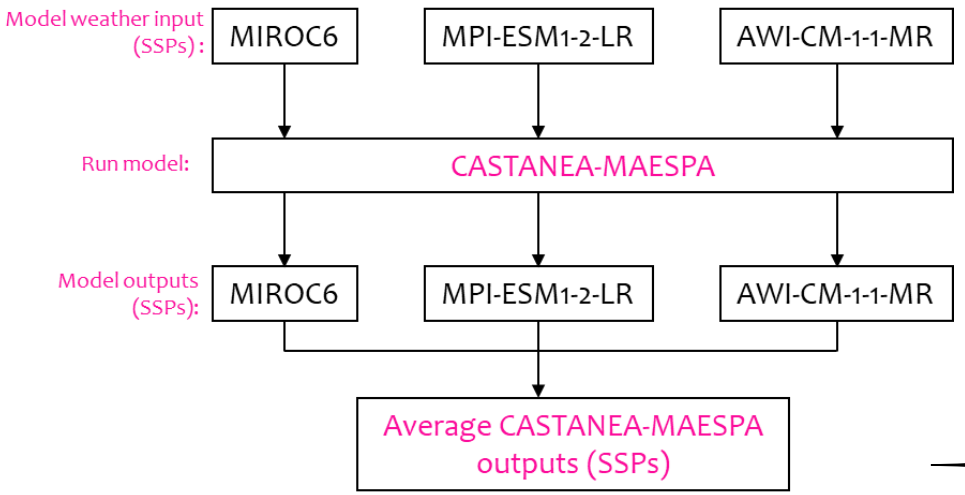


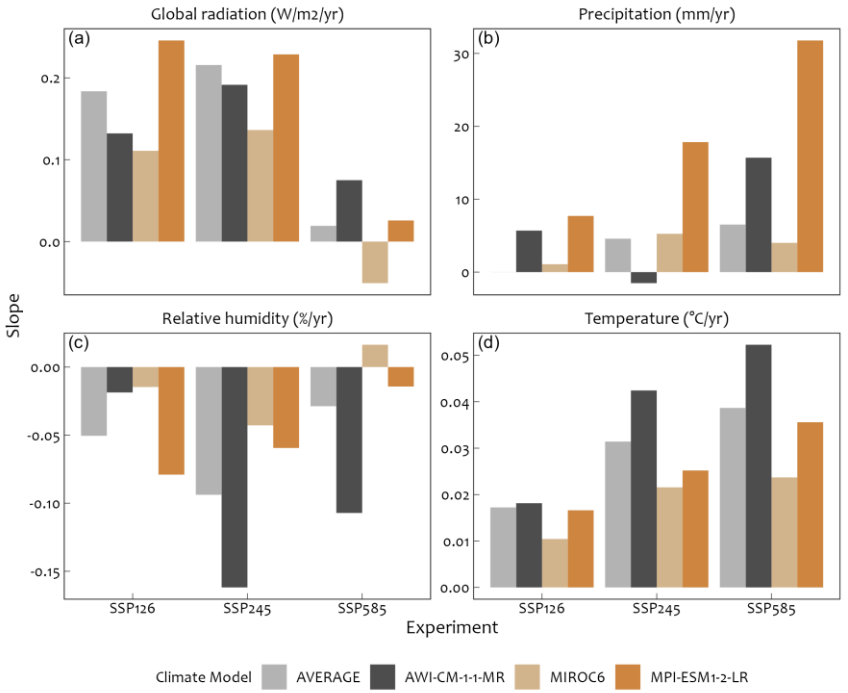
Illustration of the full carbon balance of the rubber plantation in 2018 (one year before the end of the first rotation): Simulated annual values show better agreement with measurements

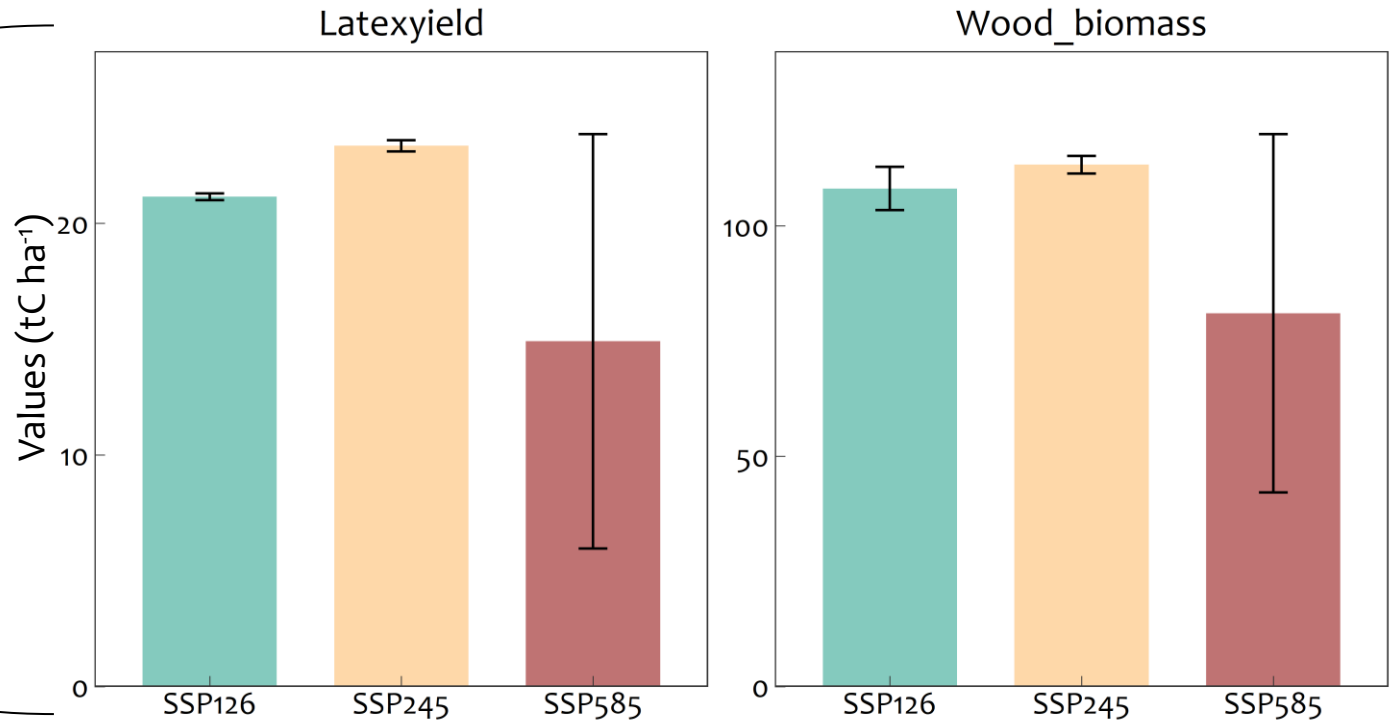
Right-angled boxes represent state variables, rounded boxes represent flow variables, and elliptical boxes represent meteorological forcing variables.

Each arrow indicates a carbon flow from one compartment to another.

Impact of different climate change scenarios on biomass and latex yield







SSP1-2.6: Sustainability - Sustainable world, low emissions
SSP2-4.5: Business as Usual - Current trends continue, moderate emissions.
SSP5-8.5: Fossil-fueled Development - Rapid economic growth, high fossil fuel use.

The overall trends suggest that, in the absence of the atmospheric CO_2 fertilization effect, higher-emission scenarios (SSP5-8.5) may lead to reduced biomass growth and latex yields

Take-home message

We showed that a process-based model can simulate key rubber tree functions and assess climate impacts.

But what is actually new ?

- adapted the CASTANEA-MAESPA model for rubber trees (and simulated a complete carbon balance).
- developed and integrated a dedicated module for simulating latex yield.
- simulations under different climate change scenarios, evaluating their potential impacts on both biomass accumulation and latex yield.

Next steps ? Improve our understanding of rubber tree phenology (e.g. effects of environmental drivers on leaf shedding, flushing, maturation, and flowering). Apply the model to other rubber plantations in Cambodia, Thailand, and Vietnam, assess climate stress impacts and provide informed recommendations…

We need more field data on phenology (LAI, litterfall), temperature responses of photosynthesis, latex yield (e.g., tapping effects), and biomass across ages to enhance model accuracy.

Thanks for your attention!

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